$$\begin{array}{c|cc} & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\ \hline \mathcal{K}^{\#1}_{0^+}\dagger & 0 & 0 \\ \mathcal{K}^{\#1}_{0^-}\dagger^{\alpha\beta\chi} & 0 & 0 \end{array}$$

$$\begin{array}{c|cccc} & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\ \hline \mathcal{K}^{\#1}_{1^+}\dagger^{\alpha\beta} & \frac{1}{6}(-\mathrm{Y}_1 k^2 + 3\overset{(2)}{\kappa}_1) & \frac{\mathrm{Y}_1 k^2 - 3\overset{(2)}{\kappa}_1}{3\sqrt{2}} & 0 & 0 \\ \mathcal{K}^{\#2}_{1^+}\dagger^{\alpha\beta} & \frac{\mathrm{Y}_1 k^2 - 3\overset{(2)}{\kappa}_1}{3\sqrt{2}} & \frac{1}{3}(-\mathrm{Y}_1 k^2 + 3\overset{(2)}{\kappa}_1) & 0 & 0 \\ \mathcal{K}^{\#1}_{1^-}\dagger^{\alpha} & 0 & 0 & \frac{1}{2}(\mathrm{Y}_2 k^2 + \overset{(2)}{\kappa}_1) & \frac{\mathrm{Y}_2 k^2 + \overset{(2)}{\kappa}_1}{\sqrt{2}} \\ \mathcal{K}^{\#2}_{1^-}\dagger^{\alpha} & 0 & 0 & \frac{\mathrm{Y}_2 k^2 + \overset{(2)}{\kappa}_1}{\sqrt{2}} & \mathrm{Y}_2 k^2 + \overset{(2)}{\kappa}_1 \end{array}$$

$$\begin{array}{c|cc} & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\ \hline \mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta} & \frac{1}{2}(3\mathrm{Y}_3 k^2 + 3\overset{(2)}{\kappa}_1) & 0 \\ \mathcal{K}^{\#1}_{2^-}\dagger^{\alpha\beta\chi} & 0 & \frac{1}{2}(3k^2\overset{(4)}{\kappa}_{15} + 3\overset{(2)}{\kappa}_1) \end{array}$$

$$\begin{array}{c|cc} & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\ \hline \mathcal{J}^{\#1}_{0^+}\dagger & 0 & 0 \\ \mathcal{J}^{\#1}_{0^-}\dagger^{\alpha\beta\chi} & 0 & 0 \end{array}$$

$$\begin{array}{c|cccc} & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\ \hline \mathcal{J}^{\#1}_{1^+}\dagger^{\alpha\beta} & \frac{1}{3\,\mathrm{Det}(1^+)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\ \mathcal{J}^{\#2}_{1^+}\dagger^{\alpha\beta} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{2}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\ \mathcal{J}^{\#1}_{1^-}\dagger^{\alpha} & 0 & 0 & \frac{1}{3\,\mathrm{Det}(1^-)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\ \mathcal{J}^{\#2}_{1^-}\dagger^{\alpha} & 0 & 0 & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{2}{3\,\mathrm{Det}(1^-)} \end{array}$$

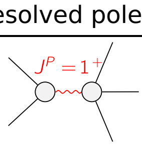
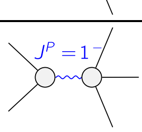
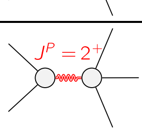
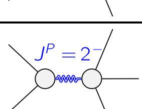
$$\begin{array}{c|cc} & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\ \hline \mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\ \mathcal{J}^{\#1}_{2^-}\dagger^{\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(2^-)} \end{array}$$

Abbreviations used in matrices

$\mathrm{Y}_1 == \overset{(4)}{\kappa}_1 - 10\overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_2 \;\&\&\; \mathrm{Y}_2 == 3\overset{(4)}{\kappa}_{15} - 2\overset{(4)}{\kappa}_2 \;\&\&\; \mathrm{Y}_3 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \;\&\&$

$\mathrm{Det}(1^+) == \frac{1}{6}(-3k^2(\overset{(4)}{\kappa}_1 - 10\overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_2) + 9\overset{(2)}{\kappa}_1) \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2}(k^2(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) + \overset{(2)}{\kappa}_1)$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$2\,\mathcal{J}^{\#1\,\alpha\beta}_{1^+} + \mathcal{J}^{\#2\,\alpha\beta}_{1^+} == 0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\chi\mathcal{J}^{\beta\alpha\chi}$
Total # constraint(s):	8	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	3	$\frac{3\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_1 - 10\overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_2}$	$-\frac{2}{\overset{(4)}{\kappa}_1 - 10\overset{(4)}{\kappa}_{15} + 7\overset{(4)}{\kappa}_2}$
	3	$\frac{\overset{(2)}{\kappa}_1}{-3\overset{(4)}{\kappa}_{15} + 2\overset{(4)}{\kappa}_2}$	$-\frac{2}{9\overset{(4)}{\kappa}_{15} - 6\overset{(4)}{\kappa}_2}$
	5	$-\frac{\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2}$	$\frac{2}{3(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2)}$
	5	$-\frac{\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_{15}}$	$-\frac{2}{3\overset{(4)}{\kappa}_{15}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	